

A CH Overcomplete Set in $C([0, \omega_1])$ with No Weak Cluster Point

arXiv automatic math run `fa_banach_001`

Candidate partial packet, June 14, 2026

Classification

- **Status:** candidate partial result.
- **Source:** T. Russo and J. Somaglia, *Overcomplete sets in non-separable Banach spaces*, arXiv:1912.08690.
- **Target:** Section 4.1 asks whether $C([0, \omega_1])$ can contain an overcomplete set with no weak cluster point, and then asks whether, say under CH, overcomplete sets can be constructed discrete in the weak topology.

Source Question

The source paper asks:

the transfinite induction and, thereby, the proof. □

One might wonder if the above result can be improved in order to produce overcomplete sets with no cluster point also in the weak topology. It is, however, elementary to realise that this is, in general, not possible. Indeed, every WLD Banach space is Lindelöf in the weak topology (see, *e.g.*, [17, Theorem 17.1]) and it is clear that every uncountable subset of a Lindelöf topological space needs to have a cluster point. On the other hand, we do not know what happens for Banach spaces that are not WLD. For example, is it possible to find an overcomplete set in $C([0, \omega_1])$ with no weak cluster point? Moreover, we do not know if it is possible, say under CH, to construct overcomplete sets that are discrete in the weak topology.

This packet proves the following CH answer for the displayed $C([0, \omega_1])$ example. It does not settle the ZFC version or the general problem for all non-WLD Banach spaces.

Definitions

For a Banach space X , a set $S \subseteq X$ with $|S| = \text{dens } X$ is *overcomplete* if every subset $\Lambda \subseteq S$ with $|\Lambda| = |S|$ is linearly dense in X .

In $K = [0, \omega_1]$, each initial segment $[0, \alpha]$, $\alpha < \omega_1$, is clopen. Hence its indicator

$$u_\alpha = \mathbf{1}_{[0, \alpha]}$$

belongs to $C(K)$.

Proof Intuition

The clopen step functions u_α have the right weak-topological shape: a cofinal pointwise limit would be $\mathbf{1}_{[0,\lambda)}$ for a limit ordinal $\lambda \leq \omega_1$, which has a jump and is not continuous. Small norm perturbations of this family still cannot have a weak cluster point in $C(K)$. Under CH, the hyperplanes of $C(K)$ can be enumerated in ω_1 steps. At stage α , choose the perturbation near u_α outside all earlier hyperplanes. Then each hyperplane contains only countably many chosen points, which is exactly the overcompleteness criterion used in Russo–Somaglia’s CH construction.

The Local Finiteness Lemma

Lemma 1. *Let $K = [0, \omega_1]$, $X = C(K)$, and fix $0 < r < 1/3$. For $\alpha < \omega_1$, put $u_\alpha = \mathbf{1}_{[0,\alpha]}$. The family of norm balls*

$$\mathcal{U} = \{B(u_\alpha, r) : \alpha < \omega_1\}$$

is locally finite for the topology of pointwise convergence on X . In particular it is locally finite for the weak topology of X .

Proof. Suppose not. Then there are $f \in C(K)$, a net of distinct ordinals (α_i) , and a net (g_i) with $g_i \in B(u_{\alpha_i}, r)$, such that g_i converges pointwise to f . Equivalently, after passing to a subnet in the compact ordinal space K , we may assume $\alpha_i \rightarrow \lambda \in K$.

Because the ordinals α_i are distinct, the subnet is not eventually constant. Therefore λ is a limit ordinal, possibly ω_1 , and

$$u_{\alpha_i}(t) \longrightarrow v_\lambda(t) \quad (t \in K), \quad v_\lambda = \mathbf{1}_{[0,\lambda)}.$$

For every $t \in K$,

$$|f(t) - v_\lambda(t)| \leq r.$$

Indeed, $g_i(t) \rightarrow f(t)$ and $\|g_i - u_{\alpha_i}\| < r$.

Now $v_\lambda(\lambda) = 0$, while $v_\lambda(\xi) = 1$ for every $\xi < \lambda$. Thus

$$|f(\lambda)| \leq r, \quad |f(\xi) - 1| \leq r \quad (\xi < \lambda).$$

Consequently, for all $\xi < \lambda$,

$$|f(\xi) - f(\lambda)| \geq 1 - 2r > 0.$$

This contradicts the continuity of f at the limit point λ . Hence $\mathcal{U}$ is pointwise locally finite. Since coordinate evaluations are weakly continuous, the same finite-coordinate neighborhood is also a weak neighborhood. $\square$

Theorem

Theorem 1 (CH). *The Banach space $C([0, \omega_1])$ contains an overcomplete set with no weak cluster point. More precisely, it contains a weakly closed-discrete overcomplete set.*

Proof. Let $X = C([0, \omega_1])$. We have $\text{dens } X = \omega_1$, and $X^* \cong \ell_1([0, \omega_1])$, so $\text{dens } X^* = \omega_1$. Under CH, $|X^*| = \omega_1$, hence the hyperplanes of X can be listed as $(H_\alpha)_{\alpha < \omega_1}$.

Fix $0 < r < 1/3$. We construct by transfinite induction a family $(x_\alpha)_{\alpha < \omega_1}$ such that

$$x_\alpha \in B(u_\alpha, r) \quad \text{and} \quad x_\alpha \notin H_\beta \quad (\beta < \alpha).$$

Assume that x_β has been chosen for $\beta < \alpha$. The ball $B(u_\alpha, r)$ is a non-empty norm-open subset of the infinite-dimensional Banach space X . It cannot be covered by countably many hyperplanes and previously chosen points. Since α is countable, we may choose

$$x_\alpha \in B(u_\alpha, r) \setminus \left(\bigcup_{\beta < \alpha} H_\beta \cup \{x_\beta : \beta < \alpha\} \right).$$

This completes the induction. Let

$$S = \{x_\alpha : \alpha < \omega_1\}.$$

We first prove overcompleteness. Let $\Lambda \subseteq \omega_1$ be uncountable. If $\overline{\text{span}}\{x_\alpha : \alpha \in \Lambda\} \neq X$, the Hahn–Banach theorem gives a hyperplane H_γ containing $\{x_\alpha : \alpha \in \Lambda\}$. But by construction

$$x_\alpha \notin H_\gamma \quad (\alpha > \gamma).$$

Thus H_γ contains at most the countably many members $\{x_\alpha : \alpha \leq \gamma\}$, contradicting the uncountability of Λ . Hence every uncountable subset of S is linearly dense in X , so S is overcomplete.

The set S has no weak cluster point because $x_\alpha \in B(u_\alpha, r)$ and the family $\{B(u_\alpha, r) : \alpha < \omega_1\}$ is weakly locally finite by the lemma. More explicitly, every $f \in X$ has a weak neighborhood that meets only finitely many of those balls, hence only finitely many points of S . Since the weak topology is Hausdorff, shrinking once more separates f from those finitely many points, except possibly from f itself. Therefore no point of X is a weak cluster point of S , and S is weakly closed-discrete. $\square$

Verification and Novelty Notes

The proof is a direct modification of Russo–Somaglia’s CH hyperplane avoidance argument, with an added localization requirement. The new ingredient is the local-finiteness lemma for small balls around the clopen initial-segment indicators. The proof uses only standard facts about $C([0, \omega_1])$: the initial segments $[0, \alpha]$ are clopen for $\alpha < \omega_1$, the dual is $\ell_1([0, \omega_1])$, and continuous functions on the ordinal compact are continuous at every countable limit and at ω_1 .

Bounded novelty check: the run indexes had no exact arXiv:1912.08690 packet or attempt. Koszmider’s later paper [2] proves the stronger background existence fact that $C([0, \omega_1])$ has overcomplete sets in ZFC, so the present packet is not novel as an existence result. Its candidate new content is the added weak-topological strengthening: under CH the overcomplete set can be chosen with no weak cluster point, indeed weakly closed-discrete. Local source search and short web phrase searches for “weak cluster point”, “weakly discrete”, and the exact $C([0, \omega_1])$ overcomplete-set question did not reveal an obvious existing answer to this strengthening.

Human Review Recommendation

The main point to check is the local-finiteness lemma. If accepted, the rest of the proof is the source paper’s CH hyperplane-avoidance construction with the additional requirement $x_\alpha \in B(u_\alpha, r)$. The result should be read as a CH affirmative answer for the highlighted $C([0, \omega_1])$ example, not as a ZFC theorem and not as a classification of all non-WLD spaces.

References

- [1] T. Russo and J. Somaglia, *Overcomplete sets in non-separable Banach spaces*, arXiv:1912.08690.
- [2] P. Koszmider, *On the existence of overcomplete sets in some classical nonseparable Banach spaces*, arXiv:2006.00806.